CCE-2

Unit 3- linked list

Title:- Call log Merger using linked list

Problem statement:

Design and implement a call log merger program in C++ using linked list. The program stores call details such as caller name and call duration in two separates call logs. it then merges the two separate call logs. it then merges the two calls and displays the merged list.

The main purpose of this program is to understand how linked lists, structures, pointers, dynamic memory allocation, insertion, traversal and merging operation are used in a peractical application.

Objectives:

Main objective:
To develop a simple call log Merger system using a linked list in c++

Specific objectives
1. To understand linked list
To understand how nodes are created and connected using pointers.

2. To store call information
   To store the caller's name and call duration in each node.
3. To perform insertion operation
   To insert new call record at the end of the linked list
4. To maintain two call logs
   To create two separate linked lists representing two different call logs.
5. To merge two linked lists
   To connect the last node of the first call log to the first node of the second call log.
6. To display call records
   To traverse the linked list and display every caller's name and duration
7. To understand pointers
   To learn how pointers are used to connect one node with another
8. To use dynamic memory allocation to create nodes dynamically during program execution
9. To develop problem-solving skills to apply data-structure concepts to a real-world problem.

Introduction

A call log is a record of telephone calls made or received by a user. it normally contains information such as the caller's name, phone number, call type, data, time, and duration. In this project, we consider only the caller name and call duration.

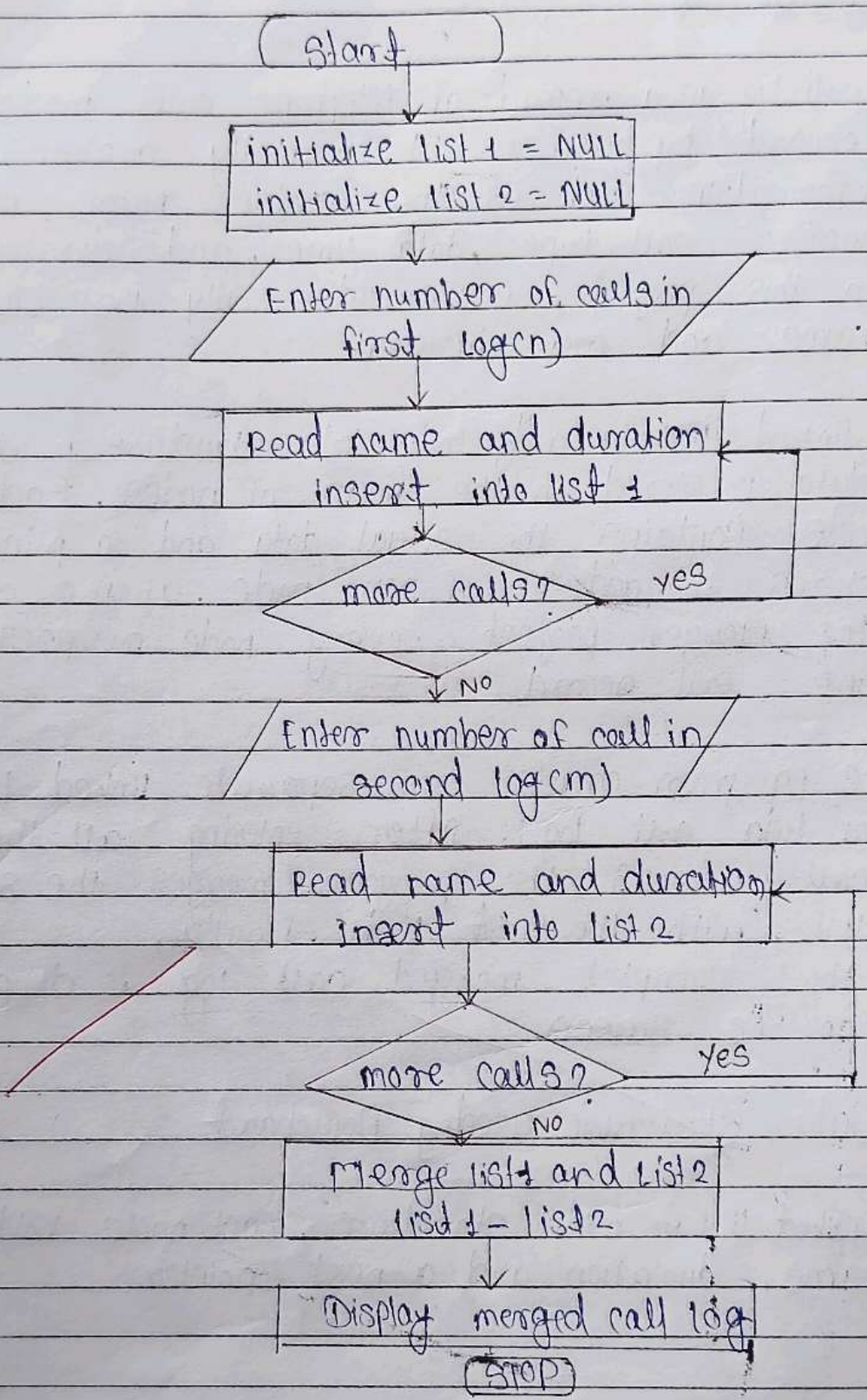
A linked list is a dynamic data structure in which data is stored in the form of nodes. Each node contains the actual data and a pointer that stores the address of next node. In the call log merger project, every node represents one call record.

The program creates two separate linked lists for two call logs. After entering all the call records, the program merges the second list with the first list. finally, the complete merged call log is displayed on the screen.

Data structure useing flowchart

Linked list in node structure, each node holds, name, duration. and a next pointer.

Flowchart


Start
initialize list 1 = NULL
initialize list 2 = NULL
Enter number of calls in first log(n)
Read name and duration insert into list 1
more calls?
yes
NO
Enter number of call in second log(m)
Read name and duration insert into list 2
more calls?
yes
NO
Merge list1 and List2
list 1 - list 2
Display merged call log
STOP

Algorithm

Step 1: Start a program.

Step 2: Create a Node structure containing:
- Caller name
- Call duration
- pointer to the next node

Step 3: Initialized two linked lists as NULL

Step 4: Ask the user to enter the number of calls in the first call log.

Step 5: Read the caller name and cell duration and insert each call at the end of the first linked list.

Step 6: Ask the user to enter the number of calls in the second call log.

Step 7: Read the caller name and call duration and insert each call at the end of the second linked list

Step 8: merge the second linked list with the first linked list by connecting the last node of the first list to the first node of the second list.

Step: 9: Display the merged call log containing all call records

Step 10: Stop the program